

Natural Language Inference using Enhanced Knowledge Graphs for Multi-Hop Reasoning

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JDSE 2026

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NLI: determine whether a hypothesis H is entailed by, neutral to, or contradicts a premise P .

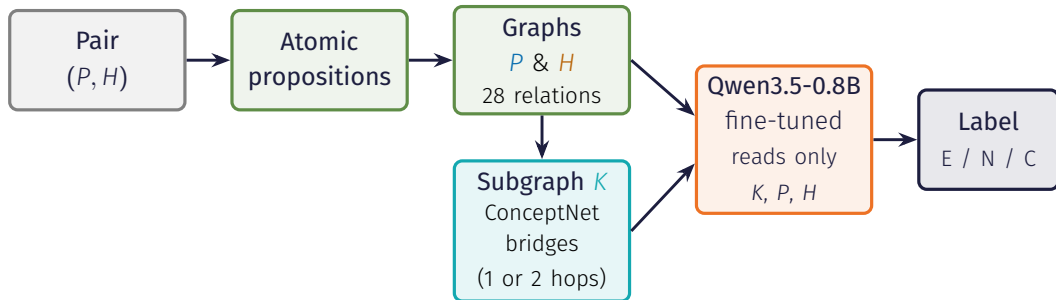
- State-of-the-art models reach over **90 %** on SNLI
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- Models exploit annotation artifacts and collapse on ANLI

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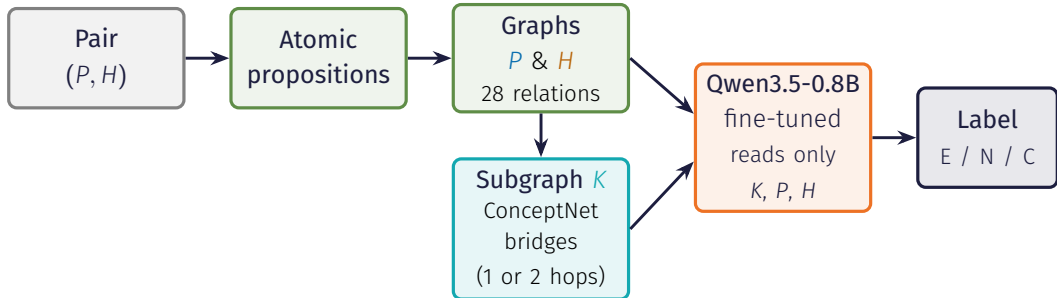
Do graphs suffice? Can a model decide from graphs only, without the text?

Pipeline



No natural language text reaches the classification stage.

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P: *Two young girls are playing soccer in a park.* **H:** *The children are outdoors.*

K: girl ISA child ; park ATLOCATION outdoors $\Rightarrow$ **entailment**

Same backbone (Qwen3.5-0.8B), same seed, same number of epochs.

	Graph	Text	Price of interp.
SNLI	89.7	91.6	−1.9
ANLI R1	56.5	70.4	−13.9
ANLI R2	46.2	57.2	−11.0
ANLI R3	46.3	55.3	−9.0

Graphs alone recover most
of the text model's accuracy.

98 % on SNLI, 80 % or more on ANLI.

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